MEASURED AGAINST PANDOC 3.13

Which pandoc format actually keeps your document

Every reader and writer pandoc ships was probed with 66 element and feature variants, one at a time. The question each probe answers: can this format tell the rich version apart from the stripped-down one at all - and if it can also be read back, does the AST come home unchanged?

77	52	66	35	3
OUTPUT FORMATS	INPUT FORMATS	ELEMENT PROBES	ROUND-TRIPPABLE	LOSSLESS WRITERS

METHOD

Two questions, two different answers

Expressiveness. For each feature a pair of documents is built: one with the feature (a table cell spanning two columns) and one without it (two plain cells). Both go through the writer. If the bytes come out identical, the format has no way to say it. This needs no reader, so it covers all 77 writers.

Round-trip identity. For the 35 formats pandoc can also read, the rich document is written and read straight back. *Exact* means the returned AST equals what went in, after normalizing `Plain`/`Para` and stripping the document wrappers ipynb, epub and fb2 add. *Canonical* is the softer verdict beside it: the content came back but the writer restyled it - a default-aligned column reported as left-aligned, a measured column width where the source gave none, raw `tex` for raw `latex`, a heading id invented by `--auto_identifiers`. Counting those as loss overstates pandoc's markdown by eleven probes, so they get their own column instead of being folded either way.

Every writer runs with `--wrap=preserve`. Without it a wrapping writer turns a SoftBreak into a space, and the probe reads as "cannot express" for a distinction the format handles fine - that alone moved 45 writers.

These measurements use released pandoc 3.13 and were refreshed on 2026-11-05. The current comparison with the preceding measurement is in `results/delta.json`; dated comparisons are in `results/history.jsonl`. Carve source, exchange AST, and export-format round trips have separate results below, with the bridge and renderer revisions recorded alongside them.

Expressiveness is the ceiling; exact round-trip is the floor. Between them sits everything that renders correctly but cannot be parsed back - which is most of it.

HEADLINE

Seven things the matrix says

63 / 62

Carve at the bridge boundary

Carve is not a pandoc format; it enters through the pandoc-carve bridge, put through the identical 66 probes. Via the exchange AST it expresses 65 features and returns 63 unchanged; via `.crv` source, 65 and 62. These scores describe the bridge boundary. Exporting through another format adds that format's losses, measured separately below.

63/66

HTML is the ceiling for real formats

html, html4/5 and every HTML-backed slide writer (revealjs, s5, slidy, slideous, dzslides) hit 63. The epub writers and chunkedhtml land one behind for different reasons: epub still folds a SoftBreak into a space, chunkedhtml spends a heading level splitting files. The three every one of them misses are latex raw inline and raw block (dropped on purpose) and the `1)` vs `1.` list delimiter. Nothing else in pandoc's AST is beyond them.

16 / 16

Attributes are the real divider

Ids, classes and key-value pairs are where formats fall apart. HTML and EPUB express 16 and 16 of 16 attribute probes, JATS 14, djot 13, pandoc markdown 12, latex 10, docx 12, typst 9, ODT 7. The attribute grid shows each format and probe.

2

Only HTML and JATS do the whole table model

Pandoc's table type has row groups, a row-head-column count, a foot, and attributes on the table, each row and each cell. Only the HTML family and JATS express all eleven table probes in their own markup - commonmark and markdown_strict also reach eleven, by falling back to a raw HTML block. LaTeX, typst, docx, ODT and asciidoc manage nine; djot and markua six; org five.

46/66

Pandoc Markdown as interchange text

It keeps *both* html and latex raw passthrough - 4/4 raw, matched only by commonmark_x, markdown_mmd and muse. It returns 46 probes exactly; html returns 44, commonmark_x 39, epub 36, and djot 35.

10

Metadata scores out of ten

markdown 10, gfm 10, and commonmark_x 10. These probes cover nested maps, booleans, custom keys, and the usual document metadata. The metadata grid gives each key's result.

0

pptx and rtf are one-way doors

The pptx writer expresses 40 features; its reader recovers none of them exactly - it is a text extractor with slide headers. rtf writes 40 and round-trips 26, dropping code, notes, blockquotes and every ordered-list style; bullets and nesting do come back.

CARVE

Where Carve lands

Carve has no pandoc reader or writer - it reaches pandoc through pandoc-carve, which maps the Pandoc AST to Carve's own AST and back. That is a different code path from a built-in reader/writer pair, but it answers the same user-facing question, so the identical 66 probes were run through it. Two lanes were measured: through `.crv` source (what `pandoc-carve report.docx -o report.crv` produces) and through the Carve exchange AST, which skips the source layer.

FORMAT	EXPRESSED/66	EXACT/66	META/10	INLINE/16	BLOCK/9	LIST/10	TABLE/11	ATTRS/16	RAW/4
carve-ast	65	63	10	16	9	9	11	16	4
carve	65	62	10	16	9	9	11	16	4
html	63	44	3.5	16	9	9	11	16	2
markdown	59	46	10	16	9	9	9	12	4
commonmark_x	57	39	10	16	8	9	8	12	4
djot	56	35	0	16	8	9	6	13	4
gfm	50	24	10	16	8	7	8	9	2

Bridge revision cf9be6e5ad29.
Renderer @markup-carve/carve 0.1.11; measured 2026-10-15T00:23:22Z.

Carve source returns 16 of 16 attribute probes, 4 of 4 raw-content probes, and 16 of 16 inline probes exactly. The grids below show which individual features survive each conversion path.

The same probes give these expressiveness and exactness scores: markdown 59 expressed / 46 exact, commonmark_x 57 / 39, djot 56 / 35, gfm 50 / 24. The Carve source lane expresses 65 features, and the AST lane 65.

Carve source round trips

42 Carve fixtures · renderer @markup-carve/carve 0.1.11 · pandoc 3.11 · bridge revision cf9be6e5ad29

CONVERSION PATH	EXACT	EQUIVALENT	RESPELLED	LOSSY	UNPARSABLE	ERR	CAME BACK READABLE/42
bridge	35	1	5	1	-	-	41
bridge-preserve	36	1	4	1	-	-	41
markdown	30	-	6	6	-	-	36
gfm	17	-	10	15	-	-	27
commonmark_x	28	-	7	7	-	-	35
djot	24	-	7	11	-	-	31
html	32	-	6	4	-	-	38
latex	15	-	6	21	-	-	21
typst	12	-	6	24	-	-	18
docx	13	1	5	23	-	-	19
odt	12	-	5	25	-	-	17
rst	18	-	6	18	-	-	24
org	18	-	7	17	-	-	25
jats	10	-	3	28	-	1	13
epub	27	-	5	10	-	-	32

These verdicts compare rendered content. A dropped comment can count as respelled because it renders to nothing; warnings record that source loss. Each format row covers writing, reading, bridge conversion, and Carve source serialization. An error may occur at any of those steps; the failure details identify the step.

`bridge-preserve` enables `roundtrip: true` metadata on the direct bridge path. Export-format rows use the default bridge options.

Table-span and table-rowspan test continuation cells; table-ragged tests a short row that Pandoc pads with an empty cell.

Preservation-mode warnings: `table-ragged`.

Carve round-trip measurement: 2026-10-15T00:23:24Z; bridge revision cf9be6e5ad290294f9387126598a5e2a009b1136.

The default bridge lane changes 7 of them: `comment` (respelled), `footnote` (respelled), `frontmatter` (equivalent), `ordered-list` (respelled), `quote-container` (respelled), `table-align` (respelled), `table-ragged` (lossy).

▶ Failures by step (including bridge failures)

The bridge says out loud what it dropped on `comment`, `table-ragged`.

What Carve does not carry, and why

- **Column widths now cross exactly.** Carve's positional `widths` metadata maps to Pandoc `ColWidth`, in both source and AST lanes.
- **Example-list numbering** (`(@)`) has no Carve form. The list is emitted as a decimal list, so the resolved numbers survive but the document-wide counter that makes separate lists renumber each other does not. Neither lane returns its document-wide counter.
- **Multiple row groups remain source-only loss.** They survive whole through `pandocToCarveAst`, while source merges body boundaries and reports that flattening. A table foot now crosses through ListTable's `footer-rows`.
- **Column alignment now has a column-level spelling.** Positional `aligns` metadata preserves it even without a header row.
- **Citations come back as citations, without their definitions.** Quoted text, citations and block cells used to change AST shape on the way back and now return unchanged. What a citation still cannot carry is the bibliography - the bridge's own warning says entries live in pandoc metadata, so it emits no `[@key]:` definitions and an undefined key renders verbatim. The probe carries no bibliography, so that part is the bridge's account, not a measurement here.

TABLE 1

Writer expressiveness, all 77 output formats

How many of the 66 features the writer can represent at all, broken out by group. Groups: 16 inline, 9 block, 10 list, 11 table, 16 attribute, 4 raw-passthrough probes. A raw score of 2/4 is normal and correct - a writer should emit raw content aimed at itself and drop raw content aimed at another target. The two highlighted rows are Carve through pandoc-carve, not pandoc built-ins.

WRITER	TOTAL/66	INLINE/16	BLOCK/9	LIST/10	TABLE/11	ATTRS/16	RAW/4	READER?
json	66	16	9	10	11	16	4	-
native	66	16	9	10	11	16	4	yes
xml	66	16	9	10	11	16	4	-
carve	65	16	9	9	11	16	4	**yes**
carve-ast	65	16	9	9	11	16	4	**yes**
dzslides	63	16	9	9	11	16	2	-
html	63	16	9	9	11	16	2	yes
html4	63	16	9	9	11	16	2	-
html5	63	16	9	9	11	16	2	-
revealjs	63	16	9	9	11	16	2	-
s5	63	16	9	9	11	16	2	-

WRITER	TOTAL/66	INLINE/16	BLOCK/9	LIST/10	TABLE/11	ATTRS/16	RAW/4	READER?
slideous	63	16	9	9	11	16	2	-
slidy	63	16	9	9	11	16	2	-
chunkedhtml	62	16	8	9	11	16	2	-
epub	62	15	9	9	11	16	2	yes
epub2	62	15	9	9	11	16	2	-
epub3	62	15	9	9	11	16	2	-
markdown	59	16	9	9	9	12	4	yes
commonmark_x	57	16	8	9	8	12	4	yes
commonmark	56	15	8	7	11	13	2	yes
djot	56	16	8	9	6	13	4	yes
beamer	55	16	8	10	9	10	2	-
docx	55	16	8	10	9	12	0	yes
jats	54	16	5	8	11	14	0	yes
jats_archiving	54	16	5	8	11	14	0	-
jats_publishing	54	16	5	8	11	14	0	-
latex	54	16	8	9	9	10	2	yes
typst	53	16	9	10	9	9	0	yes
jats_articleauthoring	52	16	5	6	11	14	0	-
markdown_strict	52	15	8	5	11	11	2	yes
context	51	16	7	9	10	9	0	-
mediawiki	51	15	8	8	9	9	2	yes
fodt	50	15	8	10	10	7	0	yes
gfm	50	16	8	7	8	9	2	yes
markdown_mmd	50	16	8	5	8	9	4	yes
odt	50	15	8	10	10	7	0	yes
opendocument	49	15	8	10	9	7	0	-
asciidoc	48	16	8	8	9	7	0	yes
asciidoc_legacy	48	16	8	8	9	7	0	-
asciidoctor	48	16	8	8	9	7	0	-
markdown_phpextra	48	15	8	5	8	10	2	yes
markua	48	16	7	9	6	10	0	-
ipynb	47	16	8	5	8	8	2	yes
rst	47	15	8	9	8	4	3	yes
markdown_github	46	15	8	5	8	8	2	yes
textile	46	13	8	8	7	7	3	yes
muse	45	15	8	7	8	3	4	yes
docbook4	44	15	6	8	7	6	2	-
org	44	15	9	7	5	5	3	yes
docbook	43	15	6	8	7	6	1	yes
docbook5	43	15	6	8	7	6	1	-
icml	42	15	8	8	9	2	0	-

WRITER	TOTAL/66	INLINE/16	BLOCK/9	LIST/10	TABLE/11	ATTRS/16	RAW/4	READER?
ms	41	15	8	8	7	3	0	-
pptx	40	15	6	8	9	2	0	yes
rtf	40	14	8	9	9	0	0	yes
tei	40	15	7	7	6	5	0	-
fb2	39	16	6	9	8	0	0	yes
ansi	38	13	8	8	7	2	0	-
dokuwiki	38	15	8	4	6	3	2	yes
texinfo	38	15	8	7	6	0	2	-
xwiki	38	13	8	4	4	7	2	-
jira	37	16	6	4	4	7	0	yes
t2t	37	15	7	4	6	2	3	yes
zimwiki	37	15	7	4	6	3	2	-
man	36	14	7	8	7	0	0	yes
haddock	35	13	6	7	9	0	0	yes
plain	35	10	5	9	9	2	0	-
bbcode_hubzilla	34	12	8	7	5	2	0	-
bbcode	33	11	8	7	5	2	0	-
vimdoc	33	8	6	9	9	1	0	-
bbcode_xenforo	32	12	8	4	5	3	0	-
bbcode_steam	30	12	8	4	5	1	0	-
bbcode_phpbb	29	11	8	7	2	1	0	-
bbcode_fluxbb	26	11	8	5	2	0	0	-
opml	1	0	1	0	0	0	0	yes
biblatex	0	0	0	0	0	0	0	-
bibtex	0	0	0	0	0	0	0	-
csljson	0	0	0	0	0	0	0	-
pdf	0	0	0	0	0	0	0	-

TABLE 2

What comes back, for the 35 formats pandoc also reads

The gap between the writer column and the exact column is the cost of using a format as storage rather than as output.
The last column lists features the writer emits correctly but the matching reader cannot recover.

FORMAT	WRITER/66	ROUND-TRIP/66	EXACT/66	CANON./66	METADATA/10	READER DROPS
native	66	66	66	-	10	-
carve-ast	65	64	63	-	10	**list_loose**
carve	65	64	62	-	10	**list_loose**
markdown	59	59	46	11	10	-

FORMAT	WRITER/66	ROUND-TRIP/66	EXACT/66	CANON./66	METADATA/10	READER DROPS
html	63	62	44	3	3.5	ol_example
commonmark_x	57	57	39	2	10	-
epub	62	61	36	2	2.5	ol_example
djot	56	56	35	-	0	-
rst	47	45	33	1	0	header_level, list_loose
muse	45	43	32	3	4.5	table_colspan, table_rowspan
asciidoc	48	47	29	-	2	header_level
markdown_mmd	50	50	28	3	6.5	-
typst	53	50	27	-	0	code_inline_attrs, list_loose, table_colwidth
markdown_phpextra	48	48	26	1	0	-
mediawiki	51	44	25	3	0	math_display, ol_style_roman, ol_style_alpha, ol_example, table_attrs, table_attrs_nocaption, table_row_attrs
docx	55	45	24	9	4	code_inline_attrs, softbreak, span_attrs, codeblock_lang, codeblock_attrs, div_attrs, ol_example, figure_attrs, table_attrs, table_attrs_nocaption
gfm	50	50	24	2	10	-
latex	54	47	24	7	4	rawblock_tex, ol_start, ol_style_roman, ol_style_alpha, ol_delim_paren, list_loose, table_colwidth
docbook	43	41	22	5	4	math_display, div_attrs
fodt	50	39	22	3	0	smallcaps, code_inline_attrs, math_display, link_title, image_size, ol_start, ol_example, list_loose, table_align, table_colwidth, table_cell_align
markdown_github	46	46	22	3	0	-
odt	50	39	22	3	0	smallcaps, code_inline_attrs, math_display, link_title, image_size, ol_start, ol_example, list_loose, table_align, table_colwidth, table_cell_align
commonmark	56	56	21	-	0	-
dokuwiki	38	35	21	1	0	rawblock_html, rawblock_tex, div_attrs
ipynb	47	66	21	3	0	-
jats	54	43	21	-	4	span_attrs, link_title, div_attrs, ol_start, ol_delim_paren, table_colwidth, table_attrs, table_attrs_nocaption, table_cell_attrs, table_row_attrs, table_rowheadcols
org	44	43	21	3	3.5	ol_delim_paren
textile	46	44	21	2	0	image_size, rawblock_tex
markdown_strict	52	52	18	-	0	-
jira	37	36	17	1	0	math_display
man	36	35	15	5	2	table_colwidth
rtf	40	26	15	1	0	code_inline, note, blockquote, codeblock, orderedlist, ol_start, ol_style_roman, ol_style_alpha, ol_delim_paren, list_loose, table_colspan, table_rowspan, table_align, tab
fb2	39	33	13	-	4	table_head, table_foot, table_colspan, table_rowspan, table_align, table_block_cell
haddock	35	32	13	1	0	list_loose, table_colspan, table_colwidth
t2t	37	35	13	-	0	rawblock_tex, div_attrs
opml	1	1	0	-	2	-
pptx	40	14	0	1	0	emph, strong, underline, strikeout, superscript, subscript, smallcaps, code_inline, math_display, linebreak, link, blockquote, codeblock, lineblock, bulletlist, orderedlist,

ipynb's round-trip column is not meaningful: its *writer* stamps a fresh UUID on every cell and the reader hands it back as a Div id, so two conversions of the same document differ. Its exact column is measured after that wrapper is stripped and is trustworthy.

TABLE 3

Which elements are hard

Sorted by how rarely a feature makes it through a round-trip intact. Nothing is universal - opml's reader rejects almost every probe - but emphasis returns exactly in 32 of the 34 readers and bullet lists in 31, where example lists and cell alignment manage one each and citations two.

ELEMENT / FEATURE	GROUP	WRITERS/74	EXACT ROUND-TRIP/34	CARVE	CARVE-AST
ol_example	list	26	0	○	○
cite	block	16	1	●	●
table_cell_align	attrs	29	1	●	●
figure_attrs	attrs	41	1	●	●
table_multibody	table	19	2	●	●
table_row_attrs	attrs	19	2	●	●
table_rowheadcols	table	22	2	●	●
codeblock_attrs	attrs	25	2	●	●
table_attrs_nocaption	attrs	25	2	●	●
table_attrs	attrs	34	2	●	●
rawblock_tex	raw	15	3	●	●
table_cell_attrs	attrs	19	3	●	●
code_inline_attrs	attrs	30	3	●	●
table_colwidth	table	45	3	●	●
table_foot	table	67	3	●	●
span_attrs	attrs	46	4	●	●
div_attrs	attrs	47	4	●	●
header_attrs	attrs	19	5	●	●
rawinline_html	raw	29	5	●	●
link_attrs	attrs	35	5	●	●
image_size	attrs	44	5	●	●
rawinline_tex	raw	10	6	●	●
table_rowspan	table	47	6	●	●
table_caption	table	66	6	●	●
table_block_cell	table	67	6	●	●
quoted_single	inline	69	6	●	●
table_colspan	table	48	7	●	●
lineblock	block	59	7	●	●
quoted_double	inline	69	7	●	●
rawblock_html	raw	32	8	●	●
ol_style_roman	list	50	8	●	●
smallcaps	inline	51	8	●	●
figure	block	65	8	●	●
ol_style_alpha	list	52	10	●	●
ol_start	list	53	10	●	●

ELEMENT / FEATURE	GROUP	WRITERS/74	EXACT ROUND-TRIP/34	CARVE	CARVE-AST
header_level	block	55	10	●	●
ol_delim_paren	list	29	11	●	●
image_title	attrs	46	11	●	●
table_align	table	55	11	●	●
header	block	69	11	●	●
orderedlist	list	69	11	●	●
link_title	attrs	32	12	●	●
math_display	inline	55	13	●	●
codeblock_lang	attrs	58	14	●	●
underline	inline	67	14	●	●
math_inline	inline	69	15	●	●
image	inline	69	17	●	●
table_head	table	67	18	●	●
deflist	list	69	18	●	●
table	table	69	18	●	●
note	block	69	21	●	●
hrule	block	61	22	●	●
subscript	inline	61	22	●	●
superscript	inline	61	22	●	●
strikeout	inline	68	22	●	●
softbreak	inline	48	23	●	●
codeblock	block	69	25	●	●
linebreak	inline	69	25	●	●
code_inline	inline	65	29	●	●
blockquote	block	67	29	●	●
link	inline	67	29	●	●
list_nested	list	69	30	●	●
list_loose	list	43	31	●	●
bulletlist	list	69	31	●	●
emph	inline	67	32	●	●
strong	inline	67	32	●	●

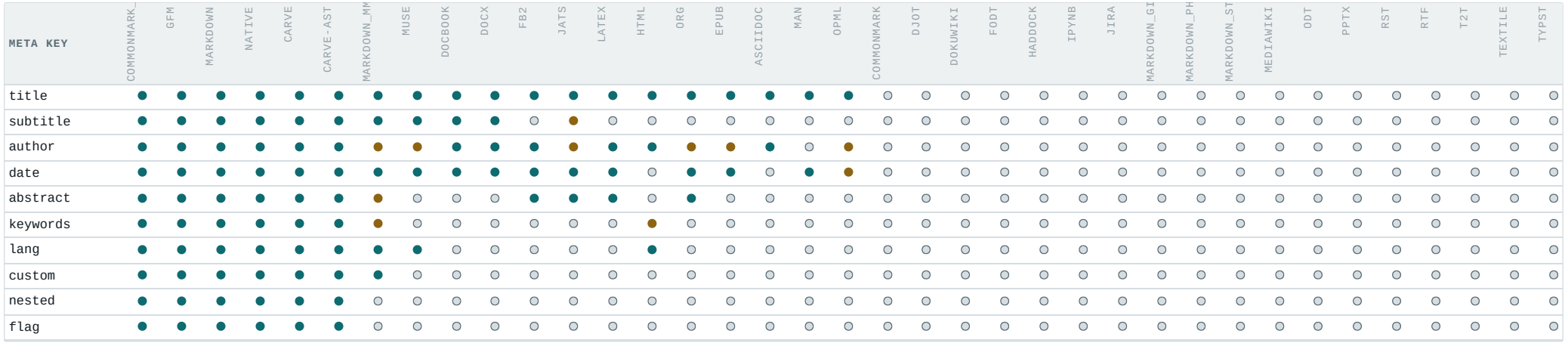
TABLE 4

Document metadata

A standalone document carrying ten metadata values, written and read back. Filled dot = identical, half dot = present but reshaped, hollow = gone.

● exact ● partial ○ lost





READING THE RESULT

Picking a target

Need everything preserved. `json`, `native` and `xml` are the AST itself and score 66/66 by construction. For anything that has to render, `html5` and `epub3` are the practical maximum.

Need it back later. `carve` through pandoc-carve returns 62 probes exactly (63 via its AST); `markdown` returns 46 and `commonmark_x` 39. For metadata, scores out of ten are: markdown 10 and Carve source 10.

Print and typesetting. `latex` 54 and `beamer` 55 lead, `typst` 53 is close and loses mostly attributes, `context` 50 has the better table story but no raw passthrough at all.

Word processing. `docx` expresses 55 probes and `odt` 50. They express 12 and 7 attribute probes respectively. `icml` keeps layout but almost no attributes.

Structured publishing. `jats` 54 is the strongest XML target and the only non-HTML writer with the complete table model; docbook and tei trail at 42 and 40.

Importing. The best sources are html, docx, latex, jats, org, rst, mediawiki and markdown. Avoid pptx and rtf as inputs when structure matters. Twelve input formats - bits, creole, csv, endnotexml, mdoc, pod, ris, tikiwiki, tsv, twiki, vimwiki, xlsx - have no matching writer and are not measured here.

Not document writers at all. `csljson`, `bibtex` and `biblatex` serialize the `references` metadata and emit nothing for body content; `opml` keeps only the heading outline. Their low scores describe their purpose, not a defect.

LIMITS

What these numbers do not say

- **Expressed is not faithful.** gfm and markdown_strict "express" colspans and tables by falling back to a raw HTML block. That counts as a difference in the output bytes, and it is genuinely lossless, but the result is no longer that format's own markup.
- **SoftBreak is a flag artifact.** Under the default `--wrap=auto` almost every text writer folds SoftBreak into a space. With `--wrap=preserve` html, markdown, latex, djot and typst all distinguish it. Treat that one row as noise.
- **Dropping foreign raw is correct.** An html writer discarding `RawBlock "latex"` is target routing working as designed, not a coverage gap. Only formats that host both (markdown, commonmark_x, markdown_mmd, muse) reach 4/4.
- **Defaults only.** No extension tuning, no `--columns`, no custom templates. Turning on `markdown_strict+raw_html` or similar moves these numbers.
- **One feature at a time.** Interactions are untested - a format may keep an id on a heading and lose it on a heading inside a table cell.
- **Carve is measured through a bridge.** The `carve` and `carve-ast` rows come from pandoc-carve's own converters, not from a pandoc reader/writer pair. Same probes, same two questions - but a library written against the Pandoc AST has an easier job than a writer constrained by an existing file format's syntax. Read the comparison as "what a user gets", not as a like-for-like implementation contest.
- **pdf scores zero because its engine is missing**, not because the format is poor. The writer shells out to an external TeX binary that is not installed here, so every pdf conversion errors and the row reads 0. Its real fidelity is LaTeX's, minus whatever the engine rasterizes away.
- **Binary formats** (docx, odt, pptx, epub, chunkedhtml) are compared after unzipping and dropping the members that carry a timestamp - `core.xml`, `app.xml`, `meta.xml`. On the path the writer lane actually uses - pandoc's server with `standalone` off - all 75 checkable writers re-emit byte-identical output for identical input, pdf excepted because it errors; `make controls` reruns that check. Standalone output is not reproducible in the same way: a CLI epub embeds a random `urn:uuid` in `content.opf` and `toc.ncx`, and ipynb a random cell id.

REPRODUCE

How it was run

```
git clone https://github.com/markup-carve/pandoc-format-fidelity
cd pandoc-format-fidelity
make                    # fetch pinned pandoc, run every lane, build the report
make carve CARVE_BRIDGE=../pandoc-carve/dist/index.js   # optional Carve lanes
```

The probe set lives in `src/probes.py`: each entry is a pair of pandoc AST block lists differing in exactly one feature. Writer conversions go through pandoc's own HTTP server for speed; the server has no filesystem access, so any media a probe references is handed over in the request; the CLI lanes get the same file through `--resource-path`. Both had to be fixed - an image pandoc cannot resolve makes epub leave the src alone and round-trip to a false *exact*.